

Yongjun Lu

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Design Engineer | Medical Devices | Prototyping | V&V | Design Controls

Education

Duke University - M.Eng. in Biomedical Engineering GPA: 3.9/4.0 Certificate: <i>Medical Device Design, Innovation and Entrepreneurship</i>	Expected May 2026 Durham, NC
Boston University - B.S. in Biomedical Engineering GPA: 3.8/4.0 Awards: <i>UROP Research Award, Dean's List</i>	Sep 2019 – May 2023 Boston, MA

Technical Skills

- **CAD and Design:** SolidWorks, Fusion 360, engineering drawings, tolerance stack-up, GD&T, design for manufacturability
- **Product Development:** Design controls, DHF documentation, design inputs/outputs, risk analysis, DFMEA/FMECA
- **Testing & Validation:** DOE, benchtop testing, test method validation, FEA, statistical analysis
- **Manufacturing & Prototyping:** 3D printing, machining, prototype fabrication, DFM, quality control
- **Programming & Robotics:** Python, MATLAB, Arduino, ROS2

Professional Experience

Brain Tool Laboratory <i>Research Technician</i>	Jun 2025 – Present Durham, NC
• Co-authored 4 technical protocols and validation reports with a PhD mentor, presenting weekly updates to Stryker engineers and supporting data-driven design and R&D improvements • Designed and iterated a custom 2-part holder to mount a surgical handpiece and depth camera on a Franka robotic arm, creating a stable and repeatable benchtop test setup for device evaluation • Developed Taguchi DOE and benchtop test methods across 3 tissue models to evaluate operating parameters of an ultrasonic aspiration system and identify optimal settings for design verification • Executed 80+ robotic resection trials using an automated workflow, troubleshooting robotic motion control and hardware integration issues to improve system reliability and reduce testing time by 50% • Built a clinical data workflow for 20+ neurosurgical cases, resolving data access and de-identification challenges to integrate clinical and benchtop datasets for performance analysis and parameter optimization	
Revairs Technology Inc. <i>Product Development Intern</i>	Jun 2024 – Nov 2024 Shanghai, China
• Conducted clinician interviews and analyzed 200+ survey responses to identify unmet needs in stress-related sleep disorders, translating insights into early-stage medical device concepts and a 30-page business plan for product development	
YCAPITAL Inc. <i>Healthcare Industry Analyst Intern</i>	Feb 2024 – Jun 2024 Shanghai, China
• Evaluated 50+ medtech companies on technology, clinical value, and market potential • Supported investment materials for a company that later secured \$2M in funding	

Design Innovation Projects

EnJoyScope - Ergonomic Colonoscope Redesign	Jan 2025 – Dec 2025
• Partnered with 2 Duke clinicians over 10 months to identify user needs related to hand fatigue, control, and usability, translating clinical feedback into design requirements for an ergonomic colonoscope handle • Led distal tip integration by designing a custom attachment and wire-guiding adaptor that enabled ~180° articulation while improving actuation stability and reducing wire tangling • Built and iterated 4+ functional prototypes, coordinating with mechanical, electronics, and user-interface subteams to align joystick input, actuation response, and distal tip motion • Explored a haptic feedback concept for the joystick with a target actuation force of 9 N to improve tactile response and support intuitive user control • Developed and executed benchtop validation tests, including 15 obstacle-navigation trials and clinician evaluations, contributing to 42% lower piercing force and 82% lower grip-force loss than a traditional endoscope • Co-developed DHF documentation, design inputs/outputs, verification plans, and uFMECA/dFMECA analyses covering 87 failure modes in alignment with FDA 21 CFR 820.30	
Insulin Syringe Design for Senior Patients	Sep 2024 – Nov 2024
• Designed a user-friendly insulin syringe with improved grip and control inspired by a pen-style clicking mechanism; developed CAD models with a thicker barrel and button-loaded injection mechanism for easier operation • Performed FEA to evaluate structural integrity under a 10 N applied force, confirming safety margins, and estimated manufacturing cost to ensure feasibility for injection molding	